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Syllabus: Description, Learning Objectives, and Grading

Hypothesis Testing

Power

Two Sample Tests

Review

Practice Quiz: Module 1

Homework (Not counted toward final grade)

30 min

Quiz: Module 1 Quiz

Submitted

🎉 Congratulations! You passed!

Grade received 92.30%

Latest Submission

Module 1 Quiz

To pass 80% or higher

Go to next item

coach

1. In a random sample of 100 patients who were taking a new drug, the mean blood pressure was 120 mmHg with a standard deviation of 4 mmHg. Assuming a one sided 5% type I error rate. The sample mean RDI had a mean of 12 and a standard deviation of 4. What value of z (testing $H_0 : \mu = x$ versus $H_a : \mu > x$) would you reject the null hypothesis?

1 / 1 point

👍 Correct

🗲 Submit your assignment

Due Jul 31, 11:59 PM IST

Attempts 3 every 8 hours

Try again

92.30%

View Feedback

We keep your highest score

2. A pharmaceutical company is interested in testing a potential blood pressure lowering medication. Their first examination considers μ as follows (SBP in mmHg):
To Pass 80% or higher

Baseline Week 2

140	138
135	136
150	148
148	146
135	133

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Test the hypothesis that there was a mean reduction in blood pressure. Compare the difference between a paired and unpaired test for a two sided 5% level test.

👍 Correct

3. Brain volumes for 9 men yielded a 90 % confidence interval of 1,077 cc to 1,123 cc. Would you reject in a two sided 5% hypothesis test of $H_0 : \mu = 1,078$?

1 / 1 point

👍 Correct

4. In an effort to improve efficiency, hospital administrators are evaluating a new triage system for their emergency room. In an validation study of the system, 5 patients were tracked in a mock (simulated) ER under the new and old triage system. Their waiting times in natural log of hours were:

Subject	1	2	3	4	5
New	0.928	-1.745	1.677	0.701	0.128
Old	2.233	-2.513	1.264	1.938	2.533

Give a Pvalue for the test of the hypothesis that the new system resulted in lower waiting times for a one sided t test.

1 / 1 point

👍 Correct

5. Refer to the previous question. Give a 95% T confidence interval for the ratio of the waiting times (recall that the measurements were natural (logged)).
Here's the data and setting again.
In an effort to improve efficiency, hospital administrators are evaluating a new triage system for their emergency room. In an validation study of the system, 5 patients were tracked in a mock (simulated) ER under the new and old triage system. Their waiting times in natural log of hours were:

Subject	1	2	3	4	5
New	0.928	-1.745	1.677	0.701	0.128
Old	2.233	-2.513	1.264	1.938	2.533

1 / 1 point

👍 Correct

6. Suppose that 18 obese subjects were randomized, 9 each, to a new diet pill and a placebo. Subjects' body mass indices (BMIs) were measured at a baseline and again after having received the treatment or placebo for four weeks. The average difference from follow-up to the baseline (followup - baseline) was -3 kg/m2 for the treated group and 1 kg/m2 for the placebo group. The corresponding standard deviations of the differences was 1.5 kg/m2 for the treatment group and 1.8 kg/m2 for the placebo group. Does the change in BMI over the two year period appear to differ between the treated and placebo groups? Assuming normality of the underlying data and a common population variance, give a pvalue for a two sided t test.

1 / 1 point

👍 Correct

7. Consider a one sided α level single group Z test of $H_0 : \mu = \mu_0$ versus $H_a : \mu > \mu_0$ with the data $\bar{X}$ for the sample mean and s for the sample standard deviation with n measurements. What are the collection of points for which you would fail to reject the hypothesis?

0 / 1 point

👎 Incorrect

8. Researchers would like to conduct a study of n healthy adults to detect a four year mean brain volume loss of $.01 \text{ mm}^3$. Assume that the standard deviation of four year volume loss in this population is $.04 \text{ mm}^3$. What would be the value of n needed for 90% power of type one error rate of 5% one sided test versus a null hypothesis of no volume loss?

1 / 1 point

👍 Correct

9. The Daily Planet ran a recent story about Kryptonite poisoning in the water supply after a recent event in Metropolis. Their usual field reporter, Clark Kent, called in sick and so Lois Lane reported the stories. Researchers plan to sample 288 individuals from Metropolis and control city Gotham and will compare mean blood Kryptonite levels (in Lex Luthors per milliliter, LL/ml). The expect to find a mean difference in LL/ml of around 2. Assuming a two sided Z test of the relevant hypothesis at 5%, what would be the power. Assume that the standard deviation is 12 for both groups.

1 / 1 point

👍 Correct

10. As you increase the type one error rate, α , what happens to power?

1 / 1 point

👍 Correct

11. Consider a setting with iid data from a $N(\mu, \sigma^2)$ distribution testing $H_0 : \mu = 0$ versus $H_a : \mu > 0$. The null hypothesis is rejected if $\sqrt{n}\bar{X}/\sigma > 1.645$. What happens to the type I error rate as n goes to infinity?

1 / 1 point

👍 Correct

12. Suppose that you have three independent samples from a $N(\mu_1, \sigma^2)$, $N(\mu_2, \sigma^2)$ and $N(\mu_3, \sigma^2)$ respectively of size n_1 , n_2 and n_3 . Let S_1^2 , S_2^2 and S_3^2 be the associated sample variances. Define the pooled variance as
$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2 + (n_3 - 1)S_3^2}{n_1 + n_2 + n_3 - 3}$$

Considering testing $H_0 : a\mu_1 + b\mu_2 + c\mu_3 = 0$. Let
$$TS = \frac{a\bar{X}_1 + b\bar{X}_2 + c\bar{X}_3}{S_p \left(\frac{a^2}{n_1} + \frac{b^2}{n_2} + \frac{c^2}{n_3} \right)^{1/2}}$$

What distribution do you think TS has under H_0 ?

1 / 1 point

👍 Correct

13. Consider a one sample Z test of $H_0: \mu = \mu_{0_05}$ versus $H_{1,a}: \mu > \mu_{0_05}$. All else equal, which scenarios will be closer to rejecting the null hypothesis?

1 / 1 point

👍 Correct